

Electrochemistry

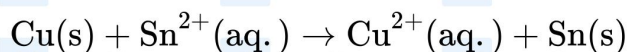
Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Q1 JEE Main 2020 - 2 September (Morning)

The Gibbs energy change (in J) for the given reaction at $[\text{Cu}^{2+}] = [\text{Sn}^{2+}] = 1\text{M}$ and 298K is:



$$(E_{\text{Sn}^{2+}}^0 = -0.16\text{V}, E_{\text{Cu}^{2+}|\text{Cu}}^0 = 0.34\text{V})$$

$$\text{Take } F = 96500\text{Cmol}^{-1})$$

Q2 JEE Main 2020 - 2 September (Evening)

For the disproportionation reaction $2\text{Cu}^+(\text{aq}) \rightleftharpoons \text{Cu(s)} + \text{Cu}^{2+}(\text{aq})$ at 298K. $\ln K$ (where K is the equilibrium constant) is $\times 10^{-1}$

Given:

$$(E_{\text{Cu}^{2+}/\text{Cu}^+}^0 = 0.16\text{V})$$

$$E_{\text{Cu}^+/\text{Cu}}^0 = 0.52\text{V}$$

$$\left(\frac{RT}{F} = 0.025\right)$$

Q3 JEE Main 2020 - 3 September (Morning)

Let C_{NaCl} and C_{BaSO_4} be the conductances (in S) measured for saturated aqueous solutions of NaCl and BaSO_4 , respectively, at a temperature T.

Which of the following is false?

(A) $C_{\text{BaSO}_4}(T_2) > C_{\text{BaSO}_4}(T_1)$ for $T_2 > T_1$

(B) $C_{\text{NaCl}}(T_2) > C_{\text{NaCl}}(T_1)$ for $T_2 > T_1$

(C) $C_{\text{NaCl}} \gg C_{\text{BaSO}_4}$ at a given T

(D) Ionic mobilities of ions from both salts increase with T.

Q4 JEE Main 2020 - 3 September (Morning)

The photoelectric current from Na (work function, $w_0 = 2.3\text{eV}$) is stopped by the output voltage of the cell



#MathBoleTohMathonGo

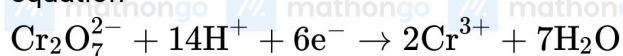
www.mathongo.com

The pH of aq. HCl required to stop the photoelectric current from K ($w_0 = 2.25\text{eV}$), all other conditions remaining the same, is $\times 10^{-2}$ (to the nearest integer).

Given, $2.303 \frac{RT}{F} = 0.06\text{V}$; $E_{\text{AgCl/Ag/Cl}^-}^0 = 0.22\text{V}$

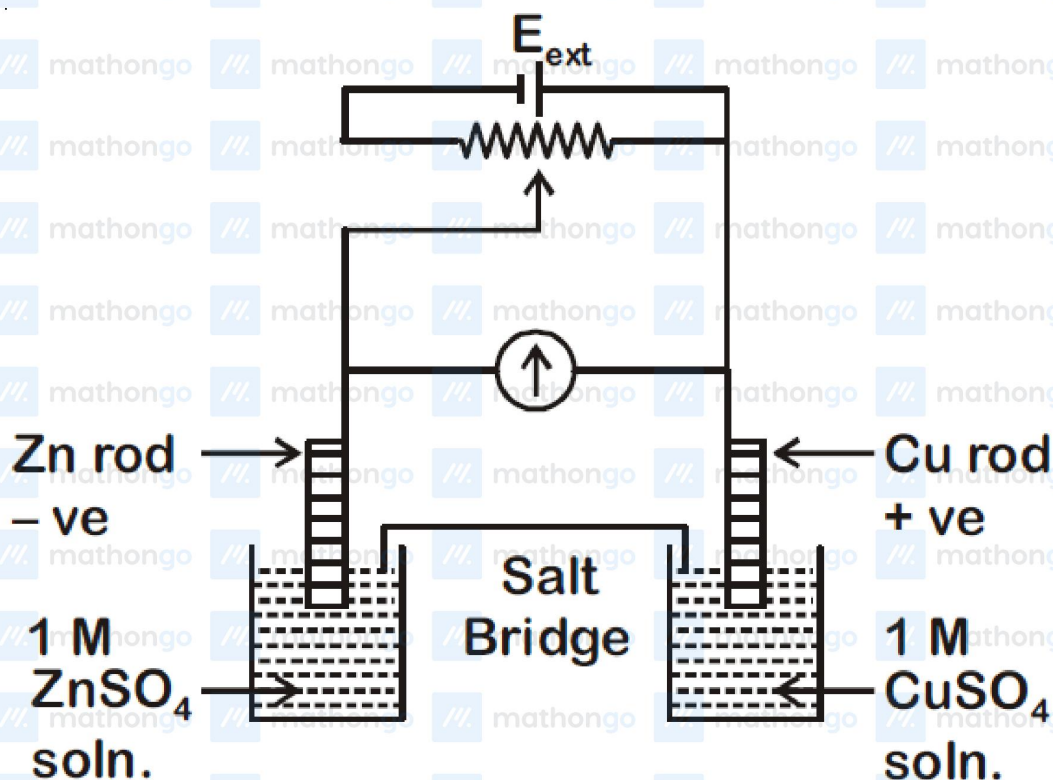
Q5 JEE Main 2020 - 3 September (Evening)

An acidic solution of dichromate is electrolyzed for 8 minutes using 2 A current. As per the following equation



The amount of Cr^{3+} obtained was 0.104g. The efficiency of the process (in%) is (Take: $F = 96000\text{C}$, At. mass of chromium = 52)

Q6 JEE Main 2020 - 4 September (Morning)



Electrochemistry

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

$$E_{\text{Cu}^{2+}|\text{Cu}}^0 = +0.34\text{V}$$

$$E_{\text{Zn}^{2+}|\text{Zn}}^0 = -0.76\text{V}$$

Identify the incorrect statement from the option below for the above cell:

- (A) If $E_{\text{ext}} < 1.1\text{V}$, Zn dissolves at anode and Cu deposits at cathode
- (B) If $E_{\text{ext}} = 1.1\text{V}$, no flow of e^- or current occurs
- (C) If $E_{\text{ext}} > 1.1\text{V}$, e^- flows from Cu to Zn
- (D) If $E_{\text{ext}} > 1.1\text{V}$, Zn dissolves at Zn electrode and Cu deposits at Cu electrode

Q7 JEE Main 2020 - 4 September (Evening)

250mL of a waste solution obtained from the workshop of a goldsmith contains 0.1M AgNO_3 and 0.1M AuCl . The solution was electrolyzed at 2V by passing a current of 1A for 15 minutes. The metal/metals electrodeposited will be

$$\left[E_{\text{Ag}^+/\text{Ag}}^0 = 0.80\text{V}, E_{\text{Au}^+/\text{Au}}^0 = 1.69\text{V} \right]$$

- (A) Silver and gold in equal mass proportion
- (B) Silver and gold in proportion to their atomic weights
- (C) Only gold
- (D) Only silver

Q8 JEE Main 2020 - 5 September (Morning)

An oxidation-reduction reaction in which 3 electrons are transferred has a ΔG^0 of 17.37kJmol^{-1} at 25°C . The value of E_{cell}^0 (in V) is $\times 10^{-2}$

$$\left(1F = 96,500\text{Cmol}^{-1} \right)$$

Q9 JEE Main 2020 - 5 September (Evening)

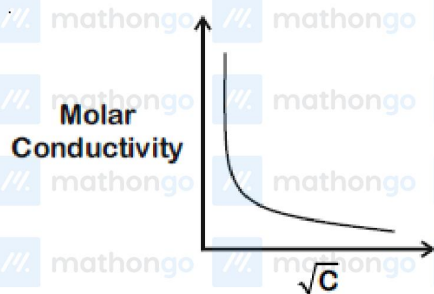
The variation of molar conductivity with concentration of an electrolyte (X) in aqueous solution is shown in the given figure.

Electrochemistry

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

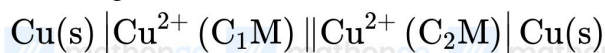


The electrolyte X is

- (A) NaCl
- (B) HCl
- (C) CH_3COOH
- (D) KNO_3

Q10 JEE Main 2020 - 6 September (Evening)

For the given cell;



change in Gibbs energy (ΔG) is negative, if

- (A) $\text{C}_2 = \sqrt{2}\text{C}_1$
- (B) $\text{C}_2 = \frac{\text{C}_1}{\sqrt{2}}$
- (C) $\text{C}_1 = 2\text{C}_2$
- (D) $\text{C}_1 = \text{C}_2$

Q11 JEE Main 2020 - 7 January (Morning)

Given that the standard potentials (E^0) of Cu^{2+}/Cu and Cu^+/Cu are 0.34 V and 0.522 V respectively, the (E^0) of $\text{Cu}^{2+}/\text{Cu}^+$ is :

- (A) 0.158 V
- (B) -0.158 V
- (C) 0.182 V
- (D) -0.182 V

Q12 JEE Main 2020 - 7 January (Evening)

#MathBoleTohMathonGo

www.mathongo.com

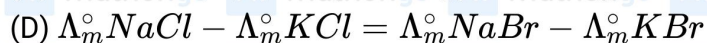
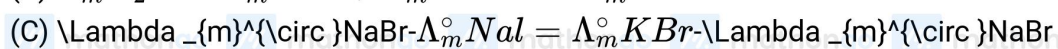
Electrochemistry

JEE Main 2020 Chapterwise

Questions with Answer Keys

Chemistry

Which of the following is incorrect?



Q13 JEE Main 2020 - 8 January (Morning)

What would be the electrode potential for the given half cell reaction at $pH = 5$?

Given : $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ $E_{\text{red}}^\circ = -1.23 \text{ V}$ ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$; Temp = 298 K ; oxygen under std.pressure of 1 bar)

Multiply your answer with -100

Q14 JEE Main 2020 - 8 January (Evening)

For an electrochemical cell $\text{Sn}(s) | \text{Sn}^{2+}(aq, 1M) || \text{Pb}^{2+}(aq, 1M) | \text{Pb}(s)$ the ratio $\frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]}$ when this cell attains equilibrium is. . . .

$$\left(E_{\text{Sn}^{2+}/\text{Sn}}^\circ = -0.14\text{V}, E_{\text{Pb}^{2+}/\text{Pb}}^\circ = -0.13\text{V}, \frac{2.303RT}{F} = 0.06 \right)$$

Multiply your answer with 100

Q15 JEE Main 2020 - 9 January (Morning)

108 g of silver (molar mass 108 g mol^{-1}) is deposited at cathode from $\text{AgNO}_3(aq)$ solution by a certain quantity of electricity. The volume (in L) of oxygen gas produced at 273 K and 1 bar pressure from water by the same quantity of electricity is. . . .

Electrochemistry

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Answer Key

Q1 (96500)

Q2 (144)

Q3 (Bonus)

Q4 (142)

Q5 (60)

Q6 (D)

Q7 (C)

Q8 (-6)

Q9 (C)

Q10 (A)

Q11 (A)

Q12 (C)

Q13 (93)

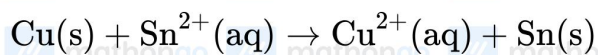
Q14 (215)

Q15 (5.67)

#MathBoleTohMathonGo

www.mathongo.com

Q1



$$E_{\text{cell}}^0 = E_{\text{Sn}^{2+}/\text{Sn}}^0 - E_{\text{Cu}^{2+}/\text{Cu}}^0$$

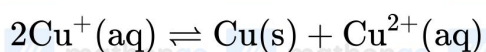
$$= -0.16 - 0.34 = -0.50\text{V}$$

$$\Delta G^0 = -nFE_{\text{cell}}^0$$

$$= -2 \times 96500(-0.50)$$

$$= 96500\text{J}$$

Q2



$$E_{\text{Cu}^{2+}/\text{Cu}^+}^0 = 0.16\text{V}; E_{\text{Cu}^+/\text{Cu}}^0 = 0.52\text{V}$$

$$E_{\text{cell}}^0 = E_{\text{Cu}^+/\text{Cu}}^0 - E_{\text{Cu}^{2+}/\text{Cu}^+}^0 = 0.52 - 0.16 = 0.36\text{V}$$

$$E_{\text{cell}}^0 = \frac{RT}{nF} \ln K$$

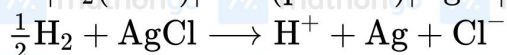
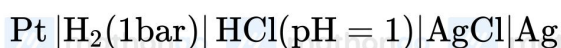
$$\ln K = \frac{0.36}{0.025} = 14.4 = 144 \times 10^{-1}$$

Q3

In general, ionic mobility increases with the increase in temperature.

Conductance (NaCl) > Conductance (BaSO₄) sparing soluble salt

Q4



$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.06}{1} \log [\text{H}^+] [\text{Cl}^-]$$

$$= 0.22 - 0.06 \log 10^{-2} = 0.34\text{V}$$

Work function of Na metal = 2.3eV

KE of photoelectron = 0.34eV

Energy of incident radiation = 2.3 + 0.34 = 2.64eV

Also energy of incident radiation for K metal = 2.64eV

Electrochemistry

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

Work function of K metal = 2.25 eV

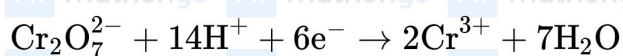
KE of photoelectrons = 2.64 - 2.25 = 0.39 eV

$$\therefore E_{\text{cell}} = 0.39 \text{ V} = 0.22 - 0.06 \log [\text{H}^+]^2$$

$$= 0.22 + 0.12 \times \text{pH}$$

$$\text{pH} \simeq 1.42 = 142 \times 10^{-2}$$

Q5



$$I = 2 \text{ A}, t = 8 \text{ min}$$

$$Q = \frac{2 \times 8 \times 60}{96000} F$$

$$\text{Moles of } \text{Cr}_2\text{O}_7^{2-} \text{ ions reduced} = \frac{960}{96000 \times 6} = \frac{1}{600}$$

$$\text{Moles of } \text{Cr}^{3+} \text{ ions obtained} = \frac{1 \times 2}{600}$$

$$\text{Mass of } \text{Cr}^{3+} \text{ ions obtained} = \frac{2 \times 52}{600} = \frac{104}{600}$$

$$\text{Mass of } \text{Cr}^{3+} \text{ ions actually obtained} = 0.104 \text{ gm}$$

$$\text{Efficiency} = \frac{6 \times 0.104 \times 100}{104} = 60\%$$

Q6

$$E_{\text{cell}}^0 = E_{\text{Cu}^{2+}/\text{Cu}}^0 - E_{\text{Zn}^{2+}/\text{Zn}}^0$$

$$= 0.34 - (-0.76)$$

$$= 1.10 \text{ V}$$

if $E_{\text{ext}} < 1.1 \text{ V}$ then Zn dissolves at anode and copper deposits

if $E_{\text{ext}} > 1.1 \text{ V}$ then Zn deposited at zinc electrodes and copper dissolves

Q7

$$\text{Millimoles of } \text{Au}^+ = 0.1 \times 250 = 25$$

$$\text{Mole of } \text{Au}^+ = \frac{25}{1000} = \frac{1}{40}$$

$$\text{Charge passed} = 1 \times t = 1 \times 15 \times 60 = 900 \text{ C}$$

$$\text{moles of } \text{e}^- \text{ passed} = \frac{900}{96500} = \frac{9}{965}$$

Only gold will be deposited as quantity of charge passed is less than the amount of Au^+ present.

#MathBoleTohMathonGo

www.mathongo.com

Electrochemistry

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

Q8

$$\Delta G^\circ = 17.37 \text{ kJ}; n = 3$$

$$\Delta G^\circ = -nFE_{\text{cell}}^\circ$$

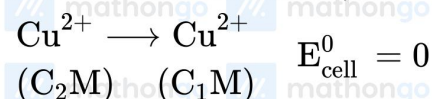
$$E_{\text{cell}}^\circ = -\frac{17.37 \times 1000}{3 \times 96500} = -0.06 = -6.00 \times 10^{-2}$$

Q9

The electrolyte (X) must be weak electrolyte. So x is CH_3COOH

Q10

For the concentration cell,



$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{2F} \ln \frac{C_1}{C_2}$$

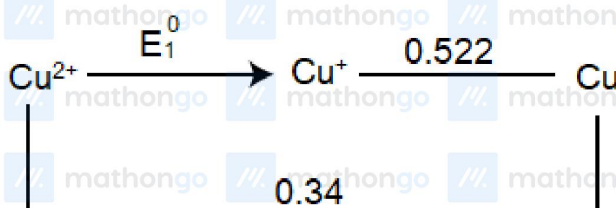
$$= \frac{RT}{2F} \ln \frac{C_2}{C_1}$$

Gibb's energy is negative only when E_{cell} is positive.

So, $C_2 > C_1$

$C_2 = \sqrt{2}C_1$ is correct.

Q11



$$2 \times 0.34 = E_1^\circ + 1 \times 0.522$$

$$E_1^\circ = 0.68 - 0.522$$

$$E_1^\circ = 0.158$$

Q12

Theory Based

#MathBoleTohMathonGo

www.mathongo.com

Electrochemistry

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

Q13

$$\begin{aligned} E &= -1.23 - \frac{0.0591}{4} \log [H^+]^4 \\ &= -1.23 + 0.0591 \times pH = -1.23 + 0.0591 \times 5 \\ &= -1.23 + 0.2955 = -0.9345 \text{ V} = -0.93 \text{ V} \end{aligned}$$

$$\text{Final ans} = -100 \times -0.93 = 93\text{V}$$

Q14

At Equilibrium state. $E_{\text{cell}} = 0$; $E^\circ_{\text{cell}} = 0.01 \text{ V}$



$$0 = 0.01 - \frac{0.06}{2} \log \left\{ \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]} \right\}$$

$$0.01 = \frac{0.06}{2} \log \left\{ \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]} \right\}$$

$$\frac{1}{3} = \log \left\{ \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]} \right\} \Rightarrow \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]} = 10^{1/3} = 2.1544$$

$$\text{Final ans} = 2.15 \times 100 = 215$$

Q15

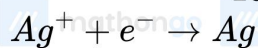
Electrochemistry

Hints & Solutions

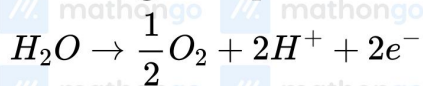
JEE Main 2020 Chapterwise

Chemistry

$$(n_{Ag})_{deposit} = \frac{108}{108} = 1 \text{ mole}$$



1 F charge is required to deposit 1 mole of Ag



2F charge deposit $\rightarrow \frac{1}{2}$ mole

1 F charge will deposit $\rightarrow \frac{1}{4}$ mole

$$V_{O_2} = \frac{nRT}{P}$$

$$= \frac{1}{4} \times \frac{0.08314 \times 273}{1} = \frac{1}{4} \times 22.7$$

$$v_{O_2} = 5.675 \text{ L}$$